$$\Gamma^{\mu}_{\nu\rho} = \frac{1}{2} g^{\mu\varepsilon} \left\{ \frac{\partial g_{\nu\varepsilon}}{\partial x^{\rho}} + \frac{\partial g_{\rho\varepsilon}}{\partial x^{\nu}} - \frac{\partial g_{\nu\rho}}{\partial x^{\varepsilon}} \right\}$$